Lesson notes · for you

Connected things, safely

IoT · Level 1 · Lesson 4

Materials
Paper and a pencil for your table; An adult at home, and the devices' boxes or manuals if you have them; A phone, to check one app's permissions

Learning goals

- Explain what makes a thing part of the Internet of Things: a sensor, a small computer and a connection.
- Give examples at home, at school and in town.
- Protect yourself: permissions, default passwords, and what you share.

Activity: Connected-things check

1. Do this with a parent or guardian. Walk round your home and list everything that connects: phones, the Wi-Fi router, a TV, a camera, a watch, a solar kit.
2. In your table, write what each one senses (if anything) and how it connects: Wi-Fi, Bluetooth, the mobile network or a cable.
3. Ask the adult whether each factory password was ever changed. Write yes, no or don't know, never the password itself.
4. On one phone, open Settings, then Apps, then one app, then Permissions.
5. Together, find one permission the app doesn't need for its job, and switch it off. Write which one and why.

Only your own family's devices, and only with an adult. Never write real passwords on your sheet, and never try to get into someone else's device or account.

Discussion

- Which connected thing at school would be most useful, and what could go wrong?
- Why do attackers want cheap devices like cameras and routers?
- What should a device never collect about children?

Check yourself

- Name the three parts of an IoT device, and one way to keep it safe.

Try it this week
With an adult, list the connected things at home, check which factory passwords were changed, and switch off one app permission that isn't needed.

Worksheet

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Name _____ Class _____ Date _____

1 **A free puzzle game asks for your microphone, your contacts and your location. What do you allow?**

- ◯ Only the location: that one is harmless
- ◯ All three, or the game won't install or run
- ◯ None: a puzzle needs none of them

2 **Which of these are part of the Internet of Things?**

- ☐ A solar calculator
- ☐ A fitness watch that sends your steps to a phone app
- ☐ A wind-up wall clock
- ☐ A tank sensor that texts the caretaker when the water is low
- ☐ A pay-as-you-go solar kit that reports to its company

3 **Nobody would bother attacking a cheap camera in a small shop, so its factory password is safe enough.**

- ◯ True
- ◯ False

4 **Put in order how a farm soil reading reaches the farmer.**

- ☐ A computer far away receives and stores it
- ☐ The farmer's phone shows an alert
- ☐ The small computer turns the reading into a short message
- ☐ The soil sensor measures that the ground is dry
- ☐ The connection sends it over the mobile network

5 **Match each risk to the best defence.**

A camera still using the password admin	• • • •	Switch off location in the camera app
A known weak spot in a router's software	• • • •	Tap Don't allow
A torch app asking for your contacts	• • • •	Switch that feature off
Photos that carry where they were taken	• • • •	Change it to a long new password
A camera microphone you never use	• • • •	Install the update

✎ **Which connected thing in your home or school worries you most, and what one change would make it safer?**

Check your answers

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1 **None: a puzzle needs none of them**

A puzzle doesn't need to hear you, read your friends' numbers or know where you are.

2 **A tank sensor that texts the caretaker when the water is low; A pay-as-you-go solar kit that reports to its company; A fitness watch that sends your steps to a phone app**

Each of the three senses something, has a small computer and sends data over a connection.

3 **False**

Attackers don't choose shops by hand. Programs like Mirai scan the internet and try factory passwords on every device they find, cheap or not.

4 **Top to bottom: 4, 5, 2, 1, 3**

Sense, decide, connect: the reading starts at the sensor, becomes a message, travels over the network and is stored before it reaches the farmer.

5 **A camera still using the password admin — Change it to a long new password**
A known weak spot in a router's software — Install the update
A torch app asking for your contacts — Tap Don't allow
Photos that carry where they were taken — Switch off location in the camera app
A camera microphone you never use — Switch that feature off

Each defence closes one door: a new password beats the attackers' lists, updates fix known weak spots, and saying no to permissions, location tags and unused features means less can leak.